

MATH 127 Calculus for the Sciences

Lecture 24

Today's lecture

Last time

Indeterminate product and indeterminate powers.

This time

Course note coverage Section 3.3.1 and 3.3.2

Asymptotes

Intervals of Increase/Decrease

Recap

Determinate form	evaluation
$\frac{1}{0^+}, \frac{\infty}{0^+}, \infty + \infty, \infty^\infty$	∞
$\frac{1}{0^-}, \frac{\infty}{0^-}$	$-\infty$
$\frac{0}{\infty}, \frac{1}{\infty}, 0^\infty$	0
Indeterminate form	tricks
$\frac{\infty}{\infty}, \frac{0}{0}$	L'Hospital's rule simplify multiply conjugate
$0 \cdot \infty$	rewrite as a quotient and apply above
$0^0, \infty^0, 1^\infty$	rewrite in base e and apply above

Horizontal asymptote

Horizontal asymptote is just another way of saying the limit at (plus or minus) infinity is whatever. Using this phrase will make you sound smart, like wearing glasses.

$y = L$ is a **horizontal asymptote** of $f(x)$



is the same as saying

$$\lim_{x \rightarrow \infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = L$$

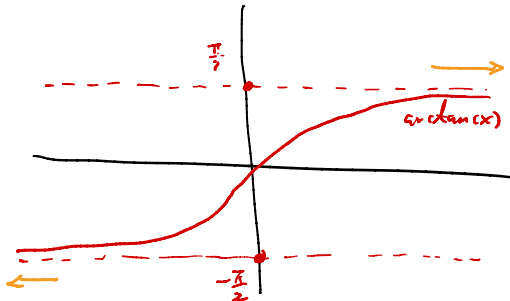
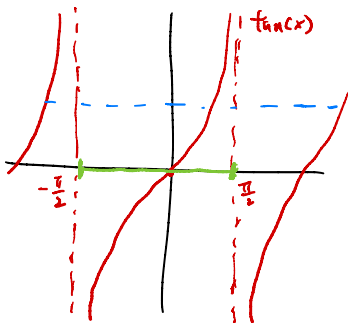
Remark Did you know the Serbian for "big brain" is "мозак"?

Horizontal asymptote

Example Consider the function $f(x) = \arctan(x)$.

- We could say the function has horizontal asymptotes given by $y = \boxed{\frac{\pi}{2}}$ and $y = \boxed{-\frac{\pi}{2}}$.
- We could also say

$$\lim_{x \rightarrow -\infty} f(x) = \boxed{-\frac{\pi}{2}}, \quad \lim_{x \rightarrow \infty} f(x) = \boxed{\frac{\pi}{2}}$$



Vertical asymptote

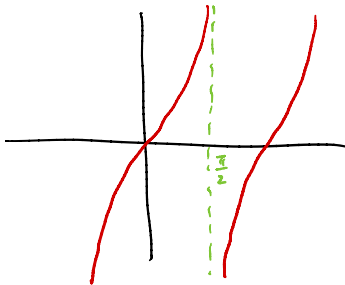
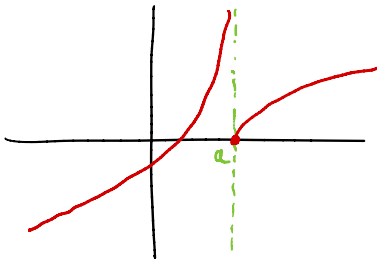
$x = a$ is a **Vertical asymptote** of $f(x)$



is the same as saying

$$\lim_{x \rightarrow a^+} f(x) = \pm\infty \quad \text{or} \quad \lim_{x \rightarrow a^-} f(x) = \pm\infty$$

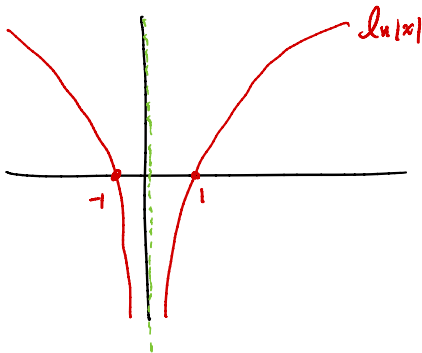
Question How are vertical asymptotes reflected on graphs?



~~Horizontal~~ asymptote

vertical.

Example Does $f(x) = \ln|x|$ have any vertical asymptotes?
Hint: Graph the function first.



By the graph, we observe
that there is a vertical asymptote
at $x=0$

Since $\lim_{x \rightarrow 0^+} \ln|x| = -\infty$

there is a vertical asymptote at $x=0$

Horizontal asymptote

$$f(x) = \frac{x}{x}$$

Example Does $f(x) = \frac{\sqrt[3]{x^3 - 2}}{(2x - 1)(x + 1)}$ have any vertical asymptotes?

Hint: this is hard to graph, but what are potential candidates for vertical asymptotes?

1. There might be ^{vertical} asymptotes when the denominator is 0

2. This happens if $(2x - 1)(x + 1) = 0$

$$2x - 1 = 0 \text{ or } x + 1 = 0$$

$$\boxed{x = \frac{1}{2}} \text{ or } \boxed{x = -1}$$



3. $\lim_{x \rightarrow \frac{1}{2}^-} f(x) = \lim_{x \rightarrow \frac{1}{2}^-} \frac{\sqrt[3]{x^3 - 2}}{(2x - 1)(x + 1)}$

This is a determinate form $\frac{L}{0^-}$ for L negative.

so the limit is ∞ or $-\infty$

$\lim_{x \rightarrow -1^+} f(x)$ is a determinate form $\frac{L}{0}$ so the limit is $\pm \infty$.

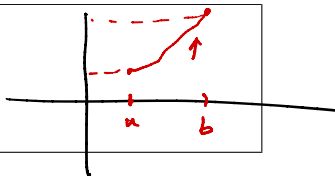
4. Hence both
 $x = \frac{1}{2}$ and $x = -1$
are vertical
asymptotes.

Interval of increase/decrease

A function $f(x)$ is **increasing** on an **interval** if

$$f(a) < f(b)$$

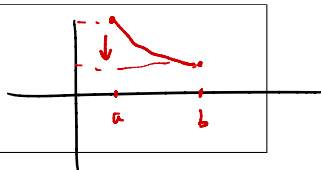
for any a, b in that **interval** such that $a < b$.



A function $f(x)$ is **decreasing** on an **interval** if

$$f(a) > f(b)$$

for any a, b in that **interval** such that $a < b$.



Derivative test

Recall that the derivative of a function measures its **rate of change**. If it has a **positive** rate of change, then the function should be **increasing**. If the rate of change is **negative**, then the function should be **decreasing**.

Theorem *The following is true:*

- $f(x)$ is **increasing** on an *interval* if

$$f'(x) > 0$$

on that interval;

- $f(x)$ is **decreasing** on an *interval* if

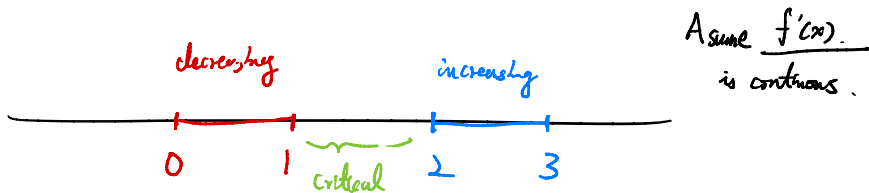
$$f'(x) < 0$$

on that interval.

Remark If $f'(a) = 0$ or does not exist, then we say a is a **critical point/critical value/critical number**.

Critical points

Question Suppose $f(x)$ is decreasing in the interval $[0, 1]$ and increasing in the interval $[2, 3]$. Does there have to be a critical point in the interval $[1, 2]$? Why or why not?



By IVT, if $f'(x) < 0$ on $[0, 1]$
 $f'(x) > 0$ on $[2, 3]$ then $f'(x) = 0$ somewhere
inside $[1, 2]$

Example: Derivative test

Example Find the intervals of increase and decrease, and the critical values of $f(x) = 3x^4 + 8x^3 + 6x^2$.

1. Find derivative $f'(x) = 12x^3 + 24x^2 + 12x$

$$\begin{aligned} &= 12 \cdot x(x^2 + 2x + 1) \\ &= 12 \cdot x \cdot (x+1)^2 \end{aligned}$$

2. Find zeroes of $f'(x)$

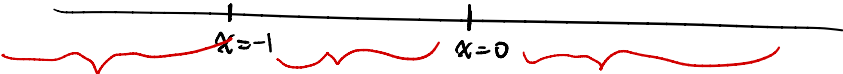
$$f'(x) = 12 \cdot x \cdot (x+1)^2 = 0$$

$$x = 0 \quad \text{or} \quad x = -1 \quad (\text{critical points})$$

3. Sketch x -axis using critical points.

Example: Derivative test

Example Find the intervals of increase and decrease, and the critical values of $f(x) = 3x^4 + 8x^3 + 6x^2$.



	$x < -1$	$-1 < x < 0$	$0 < x$
x	-	-	+
$(x+1)^2$	+	+	+
$f'(x) = 12 \cdot x \cdot (x+1)^2$	-	-	+

Conclusion $f(x)$ is decreasing on $x < -1$ and $-1 < x < 0$
increasing on $x > 0$.